Session 6: Station-Keeping and Orbit Maintenance

Task 1: Station-Keeping Dynamics

Dead-Band Control Cycle

Station-keeping maneuvers maintain a GEO satellite within an assigned station-keeping box (typically $\pm\ 0.05^o$ to $\pm\ 0.1^o$ in latitude and longitude). Rather than applying continuous low-thrust forces, station-keeping uses a limit-cycle (dead-band) approach

- **Drift Phase:** External orbital perturbations naturally force the satellite to drift across its control box.
- **Correction Burn:** When the satellite reaches the boundary limit (dead-band edge), thrusters fire impulsively to reverse the drift velocity vector, sending the satellite back toward the opposite boundary .

North-South vs. East-West Budget Dominance :

- North-South Station-Keeping (NSSK) Dominance: NSSK requires $\sim 45\ -\ 50\, m/s/yr$, accounting for over 90% of the total GEO station-keeping budget. This is driven by Third-Body gravitational perturbations (primarily Lunar-Solar gravity), which alter the orbital inclination vector away from 0° toward the ecliptic plane at an initial rate of $\sim 0.85^o/year$.
- **East-West Station-Keeping (EWSK) Economy:** EWSK requires only $\sim 2\ -\ 3\, m/s/yr$. It counters much weaker forces: Earth's geopotential triaxiality ($J_{2,2}$ tesseral harmonics causing longitudinal drift toward stable longitude wells at 75^oE and 105^oW) and Solar Radiation Pressure (SRP) affecting orbital eccentricity .

Task 2: GEO Lifetime Budget & Propellant Mass Fractions

1. 12-Year Station-Keeping Delta-V Budget

$\Delta v_{SK} = (NSSK\ +\ EWSK)\ \times\ lifetime\ =\ (48\, m/s/yr\ +\ 3m/s/yr)\ \times\ 12\, years\ =\ 612\, m$

2. End-of-Life (EOL) Graveyard Raise (+250 km)

Using the first-order mean motion approximation $n = \sqrt{\mu/a^3} \approx 7.292 \times 1^{-5}\, rad/s$ at GEO:

$$\Delta v_{gy} \approx \frac{n}{2} \cdot \Delta a = \frac{7.292\times10^{-5}}{2} \times 250000 \approx 9.115\, m/s$$

3. Total Mission Delta-V

$$\Delta v_{totla} = \Delta v_{SK} + \Delta v_{gy} = 612 + 9.115 = 621.115\, m/s$$

4. Propellant Mass Fraction Comparison ($f_p = 1 - e^{-\Delta v/(I_{sp} g_0)}$)

Taking $g_0 = 9.80665\, m/s^2$

- **Chemical Propulsion System ($I_{sp} = 300\, s$):**

$$f_p = 1 - exp(-\frac{621.115}{300\times9.80665}) = 1 - e^{-0.2111} \approx 0.1903$$

- **Ion / Electric Propulsion System ($I_{sp} = 3000\, s$):**

$$f_p = 1 - exp(-\frac{621.115}{3000\times9.80665}) = 1 - e^{-0.0211} \approx 0.0209$$

- **Trade-off Insight:** Electric propulsion reduces the propellant mass requirement by nearly 90%, saving significant space-segment mass for payload allocation at the expense of requiring higher electrical power and longer maneuver firing durations.

Task 3: LEO Reboost Analysis (450 km vs. 350 km)

- **450 km Altitude Profile**: Atmospheric density (ρ) is small enough that aerodynamic drag causes moderate orbital decay, yielding a required maintenance budget on the order of 10-20 m/s/yr to keep the spacecraft within a 5 km altitude dead-band.
- **350 km Altitude Profile**: Atmospheric density grows exponentially with decreasing altitude ($\rho(h) = \rho_0 e^{-(h-h_0)/H}$. Dropping 100 km increases thermospheric drag forces significantly, driving the annual reboost requirement up steeply to ~80-150 m/s/yr.

- **Key Trend:** Operating below 400 km introduces severe drag-penalty scaling, requiring significantly larger onboard propellant reserves or constant low-thrust continuous propulsion

Task 4: Small Maneuvers Calculations

1. Graveyard Raise (+250 km)

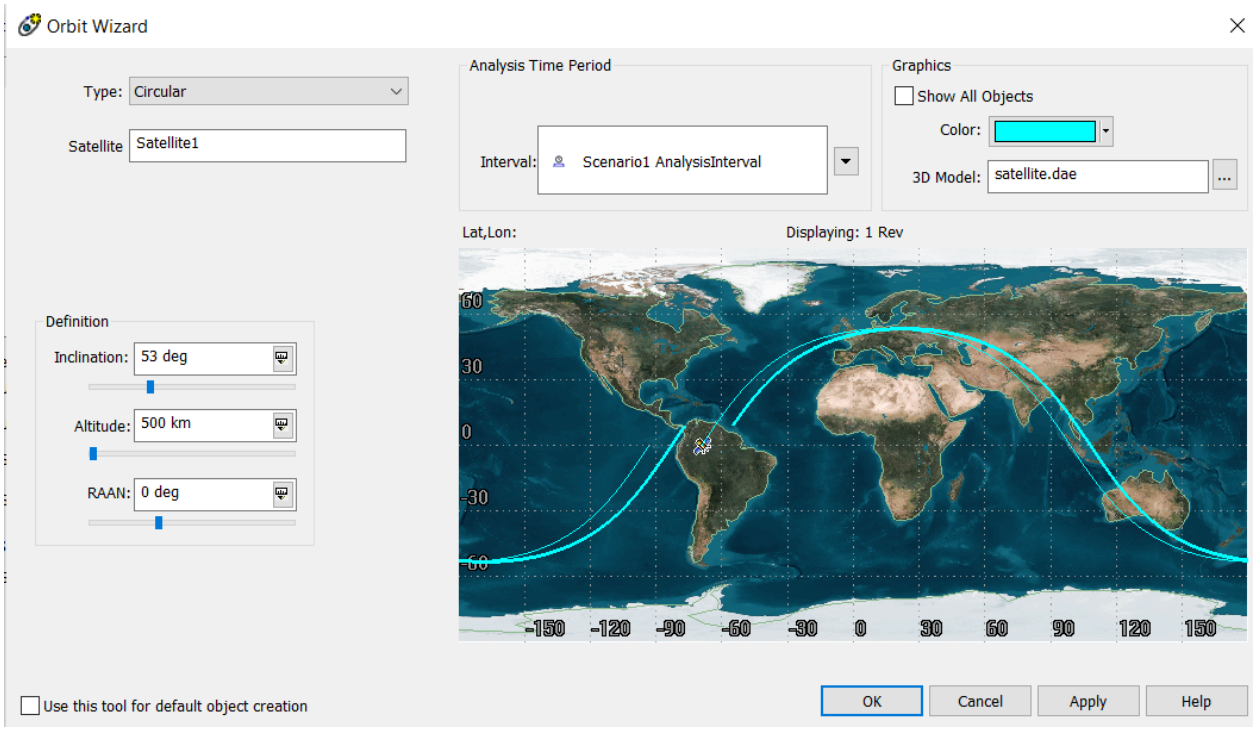
$$\Delta v_{raise} = \frac{n}{2} \cdot \Delta a = \frac{7.2921 \times 10^{-5}}{2} \times 250000 \approx 9.115\, m/s$$

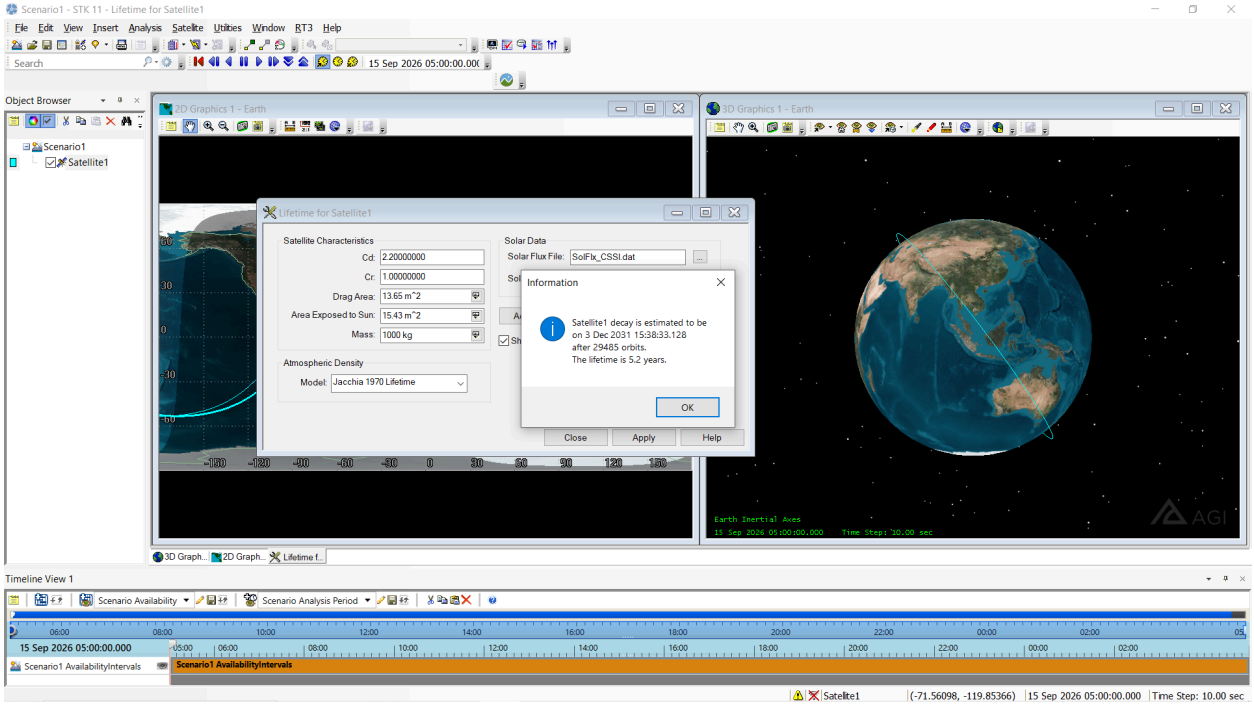
2. Inclination Correction (0.9^{o}) at GEO

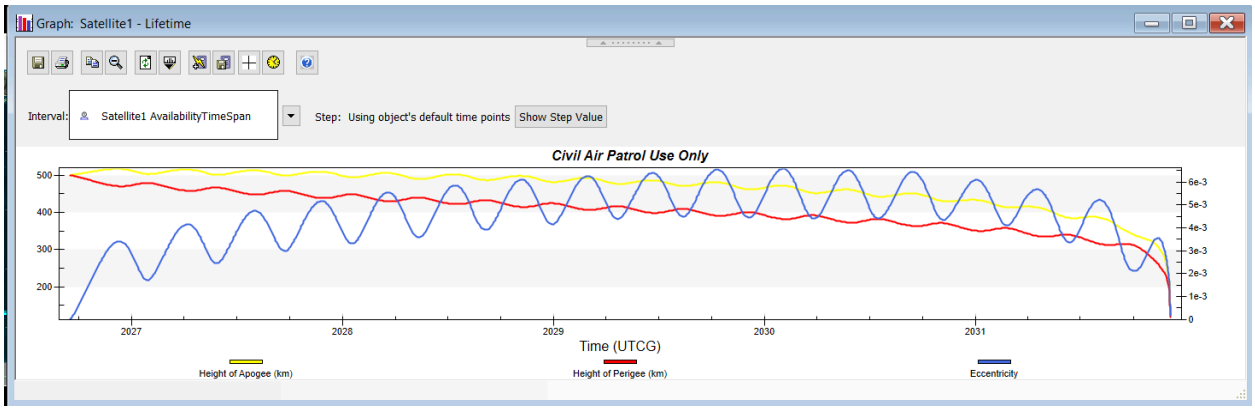
For an out-of-plane maneuver at GEO velocity $v_{GEO} = \sqrt{\mu / r_{GEO}} \approx 3.074.7\, m/s$:

$$\Delta v_{inc} = 2 \cdot v_{GEO} \cdot sin(\frac{\Delta i}{2}) = 2 \times 3074.7 \times sin(\frac{0.9^{o}}{2}) \approx 48.29\, m/s$$

Task 5: STK Lifetime Tool




Scenario1 - STK 11 - Lifetime for Satellite1
File Edit View Insert Analysis Satellite Utilities Window RT3 Help
Object Browser
Scenario1
Satellite1
2D Graphics 1 - Earth
3D Graphics 1 - Earth
Lifetime for Satellite1
Satellite Characteristics
Cd: 2.20000000
Cr: 1.00000000
Drag Area: 13.65 m^2
Area Exposed to Sun: 15.43 m^2
Mass: 1000 kg
Atmospheric Density
Model: Jacchia 1970 Lifetime
Solar Data
Solar Flux File: SolFlx_CSSI.dat
Information
Satellite1 decay is estimated to be on 3 Dec 2031 15:38:33.128 after 29485 orbits.
The lifetime is 5.2 years.
OK
Close
Apply
Help
Timeline View 1
Scenario Availability
Scenario Analysis Period
15 Sep 2026 05:00:00.000
Scenario1 AvailabilityIntervals
Scenario1 AvailabilityIntervals
Satellite1
(-71.56098, -119.85366)
15 Sep 2026 05:00:00.000
Time Step: 10.00 sec


Graph: Satellite1 - Lifetime
Interval: Satellite1 AvailabilityTimeSpan
Step: Using object's default time points
Show Step Value
Civil Air Patrol Use Only
500
400
300
200
6e-3
5e-3
4e-3
3e-3
2e-3
1e-3
0
2027
2028
2029
2030
2031
Time (UTCG)
Height of Apogee (km)
Height of Perigee (km)
Eccentricity